Grade 5 Math — Fractions Unit (Spring)

Teacher: Ms. Alvarez · 4 weeks · Before this unit: whole-number multiplication and division

Unit goal

Students can explain *why* fraction rules work using pictures, number lines and words, not just follow steps.

Part 1 — Understanding Fractions

1. **What a Fraction Is.** A fraction names equal parts of a whole; the numerator counts parts, the denominator says how many equal parts make the whole. *Mastery:* explain what the numerator and denominator mean using a picture of equal parts. 2. **Fractions on a Number Line.** Placing fractions between 0 and 1 and beyond 1 by splitting each whole into equal jumps. *Mastery:* place a fraction on a number line and explain how the equal jumps were chosen. *Requires:* What a Fraction Is. 3. **Equivalent Fractions.** Different fractions can name the same amount; multiplying or dividing top and bottom by the same number. *Mastery:* show that two fractions are equivalent with a picture and explain why multiplying top and bottom by the same number keeps the amount the same. *Requires:* What a Fraction Is.

Part 2 — Comparing and Combining

4. **Comparing Fractions.** Deciding which fraction is bigger using common denominators, common numerators, or benchmarks like one half. *Mastery:* compare two fractions and justify the answer with a benchmark or a common denominator. *Requires:* Equivalent Fractions, Fractions on a Number Line. 5. **Adding and Subtracting Like Fractions.** Adding or subtracting parts of the same size; the denominator stays the same. *Mastery:* explain why the denominator does not change when adding fractions with the same denominator. *Requires:* What a Fraction Is. 6. **Adding Unlike Fractions.** Rewriting fractions with a common denominator before adding or subtracting. *Mastery:* add two fractions with different denominators and explain why the pieces must be the same size first. *Requires:* Equivalent Fractions, Adding and Subtracting Like Fractions.

Part 3 — Beyond One Whole

7. **Mixed Numbers.** Writing amounts greater than one as a whole number and a fraction, and converting to and from improper fractions. *Mastery:* convert between a mixed number and an improper fraction and explain it with a picture. *Requires:* Fractions on a Number Line. 8. **Fractions and Decimals.** Tenths and hundredths as fractions and decimals; reading money and

measurements. *Mastery:* write a fraction with denominator 10 or 100 as a decimal and explain what each digit means. *Requires:* Equivalent Fractions.